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Anderson

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(54) **METHOD FOR LOCATING AND PLANTING SENTINEL PLANTS**

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CPC **A01C 14/00** (2013.01); **A01B 69/008** (2013.01); **A01C 5/04** (2013.01); **G05D 1/0094** (2013.01)

(58) **Field of Classification Search**

CPC **A01C 14/00**
See application file for complete search history.

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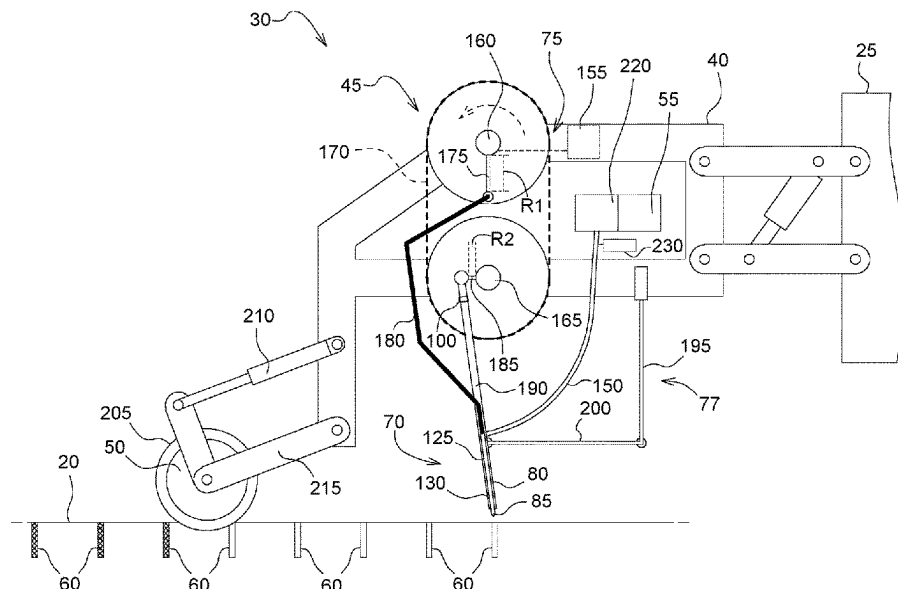
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ABSTRACT

A planter configured for planting a sentinel plant in a field. The planter comprises a first planting system that is coupled to the planter. The first planting system is configured for planting a first sentinel plant having a first sentinel plant characteristic. A global positioning system is communicatively coupled to the planter. The global positioning system is configured for generating a location signal indicative of a location of the planter. A control system is communicatively coupled to the planter. The control system is configured to receive the first sentinel plant characteristic, receive the location signal, receive a georeferenced field characteristic, and control the first planting system based on at least one of the first sentinel plant characteristic, the location, or the georeferenced field characteristic.

14 Claims, 7 Drawing Sheets



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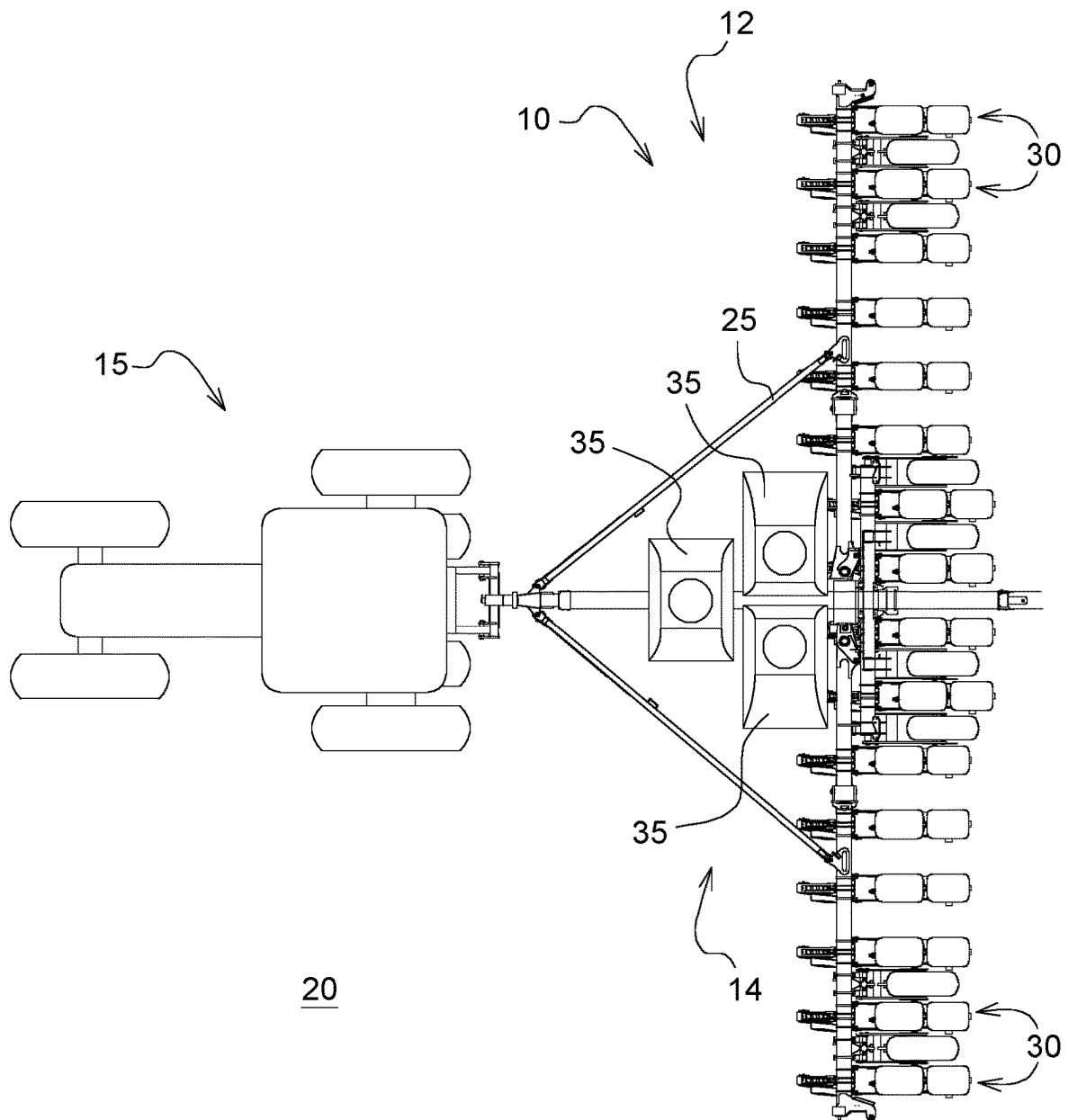
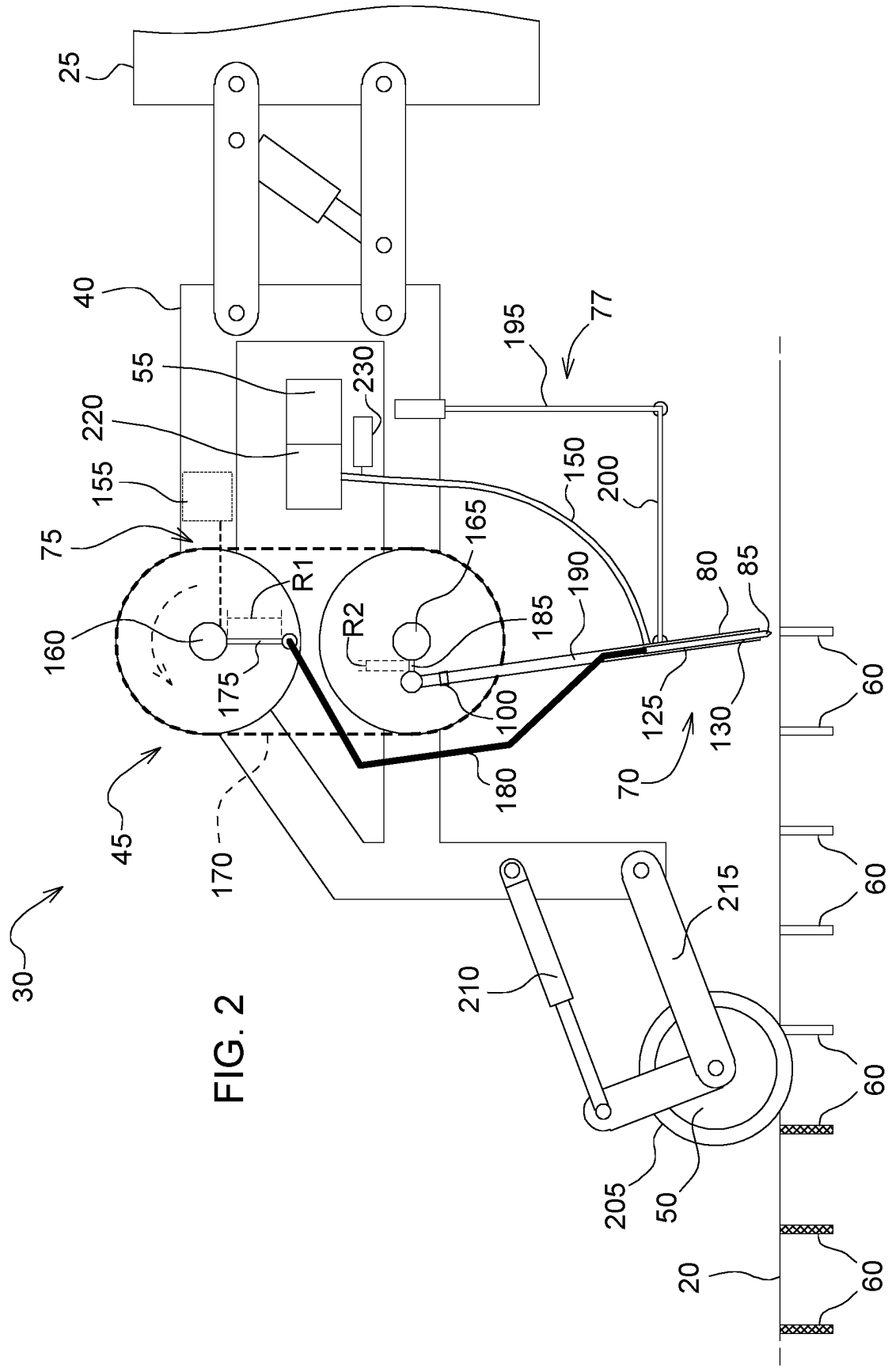


FIG. 1



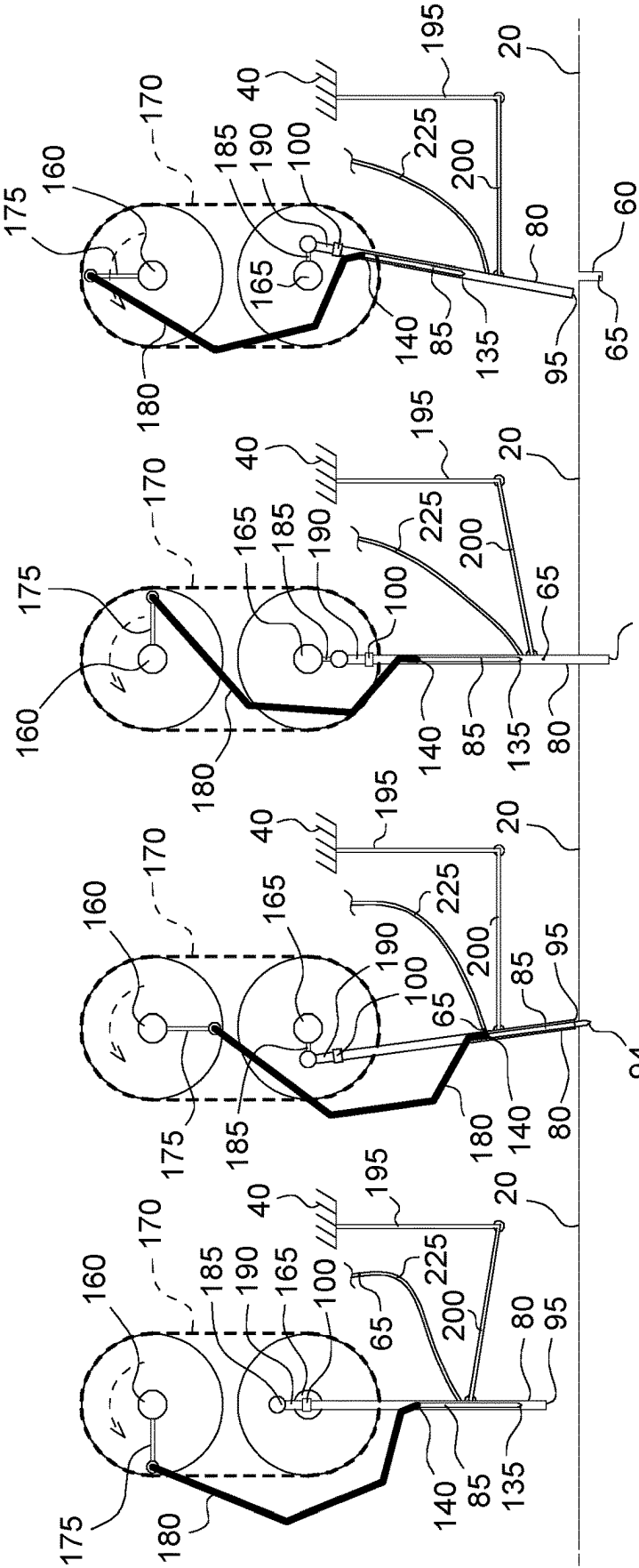
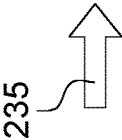


FIG. 3

FIG. 4

FIG. 5

FIG. 6

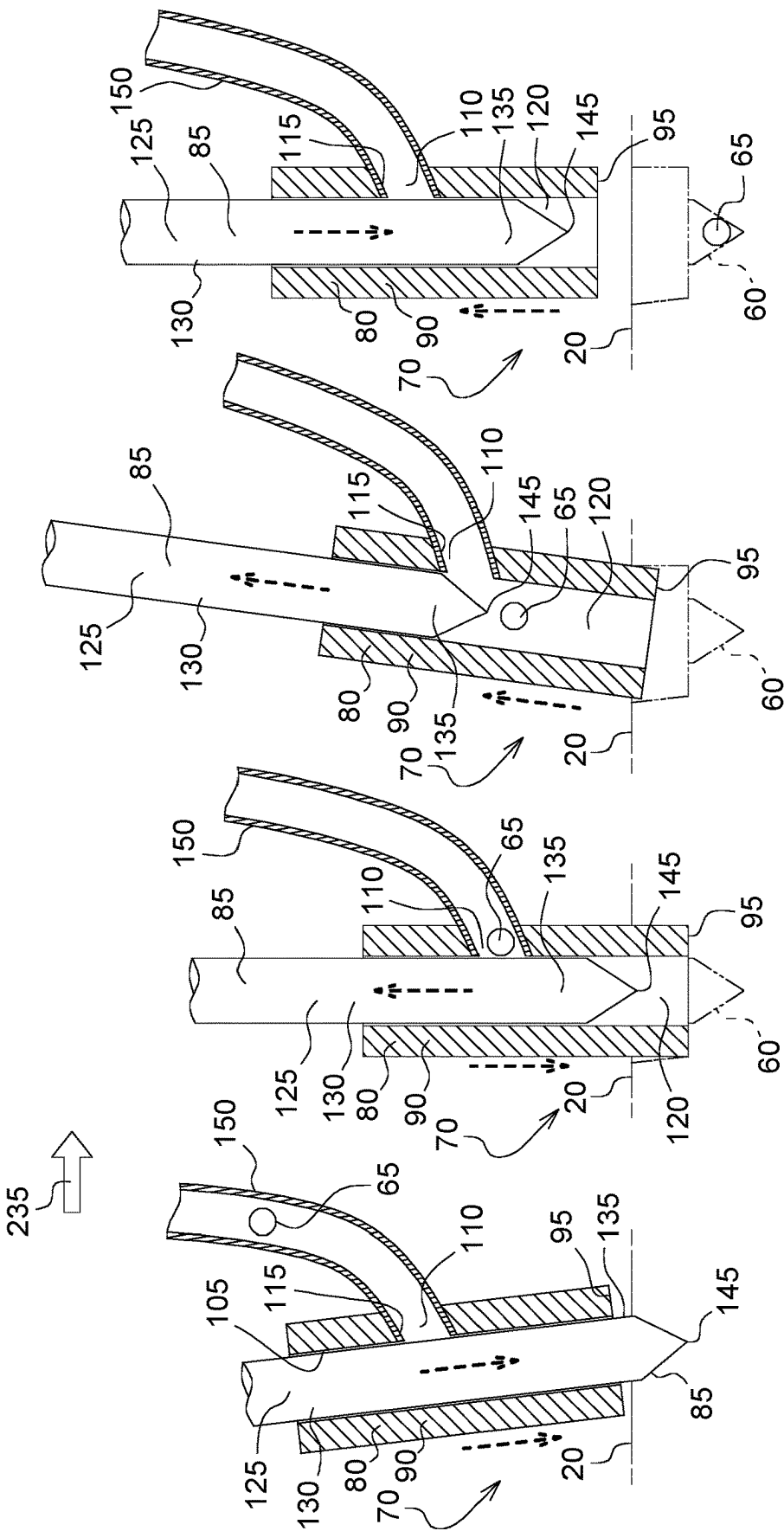
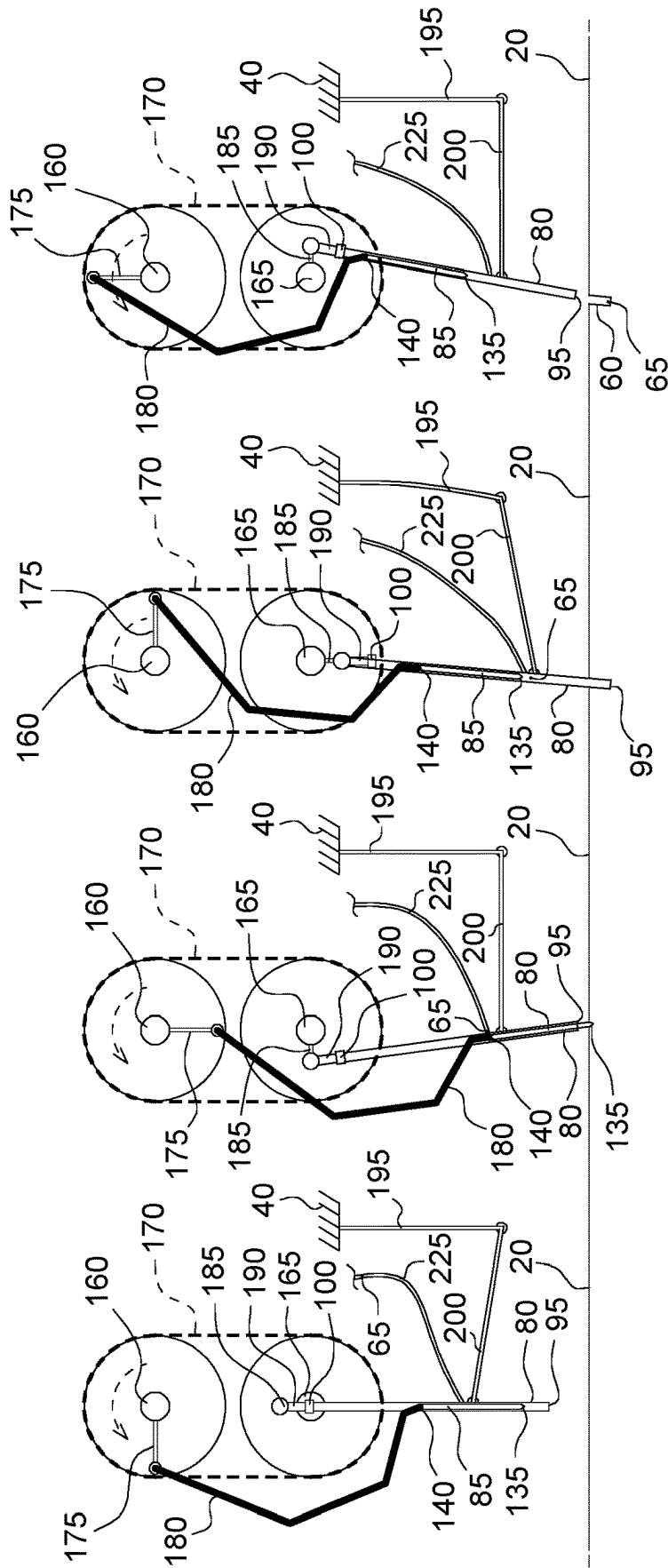
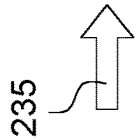


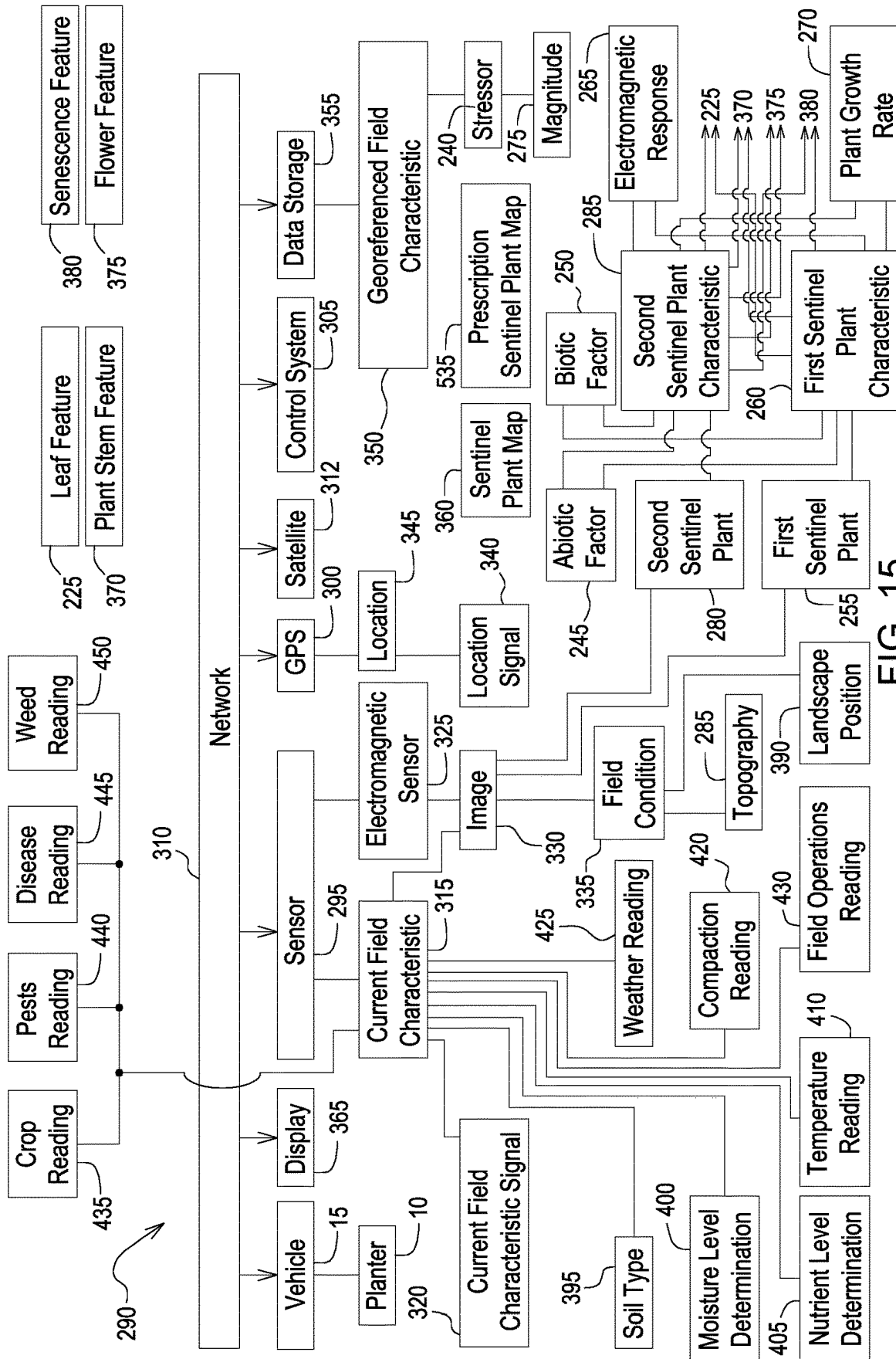
FIG. 10

FIG. 9

FIG. 8

FIG. 7





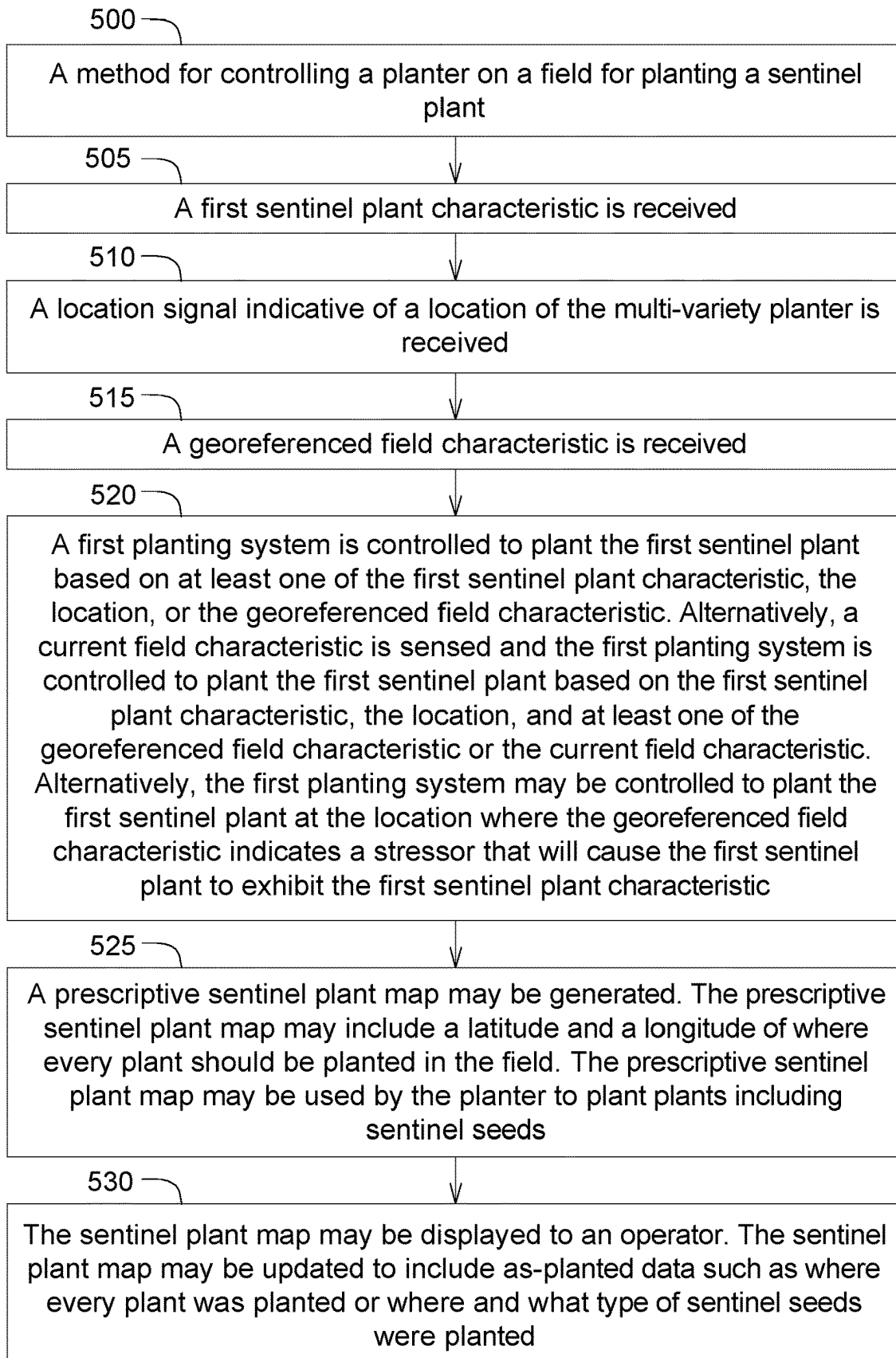


FIG. 16

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METHOD FOR LOCATING AND PLANTING SENTINEL PLANTS

FIELD OF THE DISCLOSURE

The present disclosure generally relates to a planter and more particularly to a planter and a method for planting sentinel plants in a field.

BACKGROUND OF THE DISCLOSURE

Planters are generally used to plant plants in a field by delivering a seed from a hopper to the ground with a seed delivery system. Multi-variety planters can simultaneously plant seeds from two or more different plant genotypes in a single planting operation.

SUMMARY OF THE DISCLOSURE

In one embodiment, a planter configured for planting a sentinel plant in a field is disclosed. The planter comprises a first planting system that is coupled to the planter. The first planting system is configured for planting a first sentinel plant having a first sentinel plant characteristic. A global positioning system is communicatively coupled to the planter. The global positioning system is configured for generating a location signal indicative of a location of the planter. A control system is communicatively coupled to the planter. The control system is configured to receive the first sentinel plant characteristic, receive the location signal, receive a georeferenced field characteristic, and control the first planting system based on at least one of the first sentinel plant characteristic, the location, or the georeferenced field characteristic.

In another embodiment, a multi-variety planter configured for planting a sentinel plant in a field is disclosed. The multi-variety planter comprises a first planting system that is coupled to the multi-variety planter. The first planting system is configured for planting a first sentinel plant having a first sentinel plant characteristic. A second planting system is coupled to the multi-variety planter. The second planting system is configured for planting a second sentinel plant having a second sentinel plant characteristic. A global positioning system is communicatively coupled to the multi-variety planter. The global positioning system is configured for generating a location signal indicative of a location of the multi-variety planter. A control system is communicatively coupled to the multi-variety planter. The control system is configured to receive the first sentinel plant characteristic, receive the second sentinel plant characteristic, receive the location signal, receive a georeferenced field characteristic, and control at least one of the first planting system or the second planting system based on at least one of the first sentinel plant characteristic, the second sentinel plant characteristic, the location, or the georeferenced field characteristic.

In yet another embodiment, a method of controlling a planter for planting a sentinel plant in a field is disclosed. The method comprises receiving a first sentinel plant characteristic. Receiving a location signal indicative of a location of the planter. Receiving a georeferenced field characteristic. Controlling a first planting system to plant the first sentinel plant based on at least one of the first sentinel plant characteristic, the location, or the georeferenced field characteristic.

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Other features and aspects will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a top schematic view of a tractor and a planter having a plurality of row units;

FIG. 2 is a side view of one of the plurality of row units of the planter of FIG. 1;

FIGS. 3-6 are schematic side views of a portion of the row unit of FIG. 2 cycling through a series of positions;

FIGS. 7-10 are schematic side views of a planting assembly of the row unit of FIG. 2 cycling through a series of positions;

FIGS. 11-14 are schematic side views of a portion of the row unit of FIG. 2 cycling through a series of positions under alternate conditions;

FIG. 15 is a block diagram of a planter; and

FIG. 16 is a flow diagram of a method for operating a planter in a field.

Before any embodiments are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Further embodiments of the invention may include any combination of features from one or more dependent claims, and such features may be incorporated, collectively or separately, into any independent claim.

As used herein, unless otherwise limited or modified, lists with elements that are separated by conjunctive terms (e.g., "and") and that are also preceded by the phrase "at least one of" or "one or more of" indicate configurations or arrangements that potentially include individual elements of the list, or any combination thereof. For example, "at least one of A, B, and/or C" or "one or more of A, B, and/or C" indicates the possibilities of only A, only B, only C, or any combination of two or more of A, B, and C (e.g., A and B; B and C; A and C; or A, B, and C).

DETAILED DESCRIPTION

FIG. 1 illustrates a planter 10 for planting plants 11 (FIG. 6) in seed form including sentinel seeds 65 (FIG. 6). The planter 10 may be a high-speed planter, an in-furrow planter, a multi-variety planter, a seedling planter, an air seeder, a broadcast seeder, or other type of planter. A first planting system 12 for planting one variety of plant 11 in the form of a seed may be coupled to the planter 10 or the planter 10 may be a multi-variety planter and also have a second planting system 14 for planting more than one variety of plant 11 in the form of a seed. The planter 10 for planting seeds is disclosed in commonly assigned U.S. patent application Ser. No. 17/107,041, the entire contents of which are hereby incorporated by reference. Other implementations of the first planting system 12 or the second planting system 14 for planting plants 11 in seedling form are contemplated by this disclosure such as that disclosed in commonly assigned U.S. patent application Ser. No. 16/511,918, the entire contents of which are hereby incorporated by reference.

The illustrated planter 10 is selectively coupled to a vehicle 15, such as a tractor, to be pulled across a field 20 or ground (FIG. 2). The planter 10 includes a planter frame 25. The illustrated first planting system 12 and second planting system 14 each include at least one planting row unit 30 and

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at least one hopper **35** containing seeds to be planted, which are coupled to the planter frame **25**. In some implementations the seeds are all the same while in other implementations, the seeds may be different by hopper **35** and may be planted differently by row or by location within a row. In other examples, without limitation, the planter frame **25** may be rigidly attached to a prime mover such as with a three-point hitch of vehicle **15** or planter frame **25** may be permanently integrated with vehicle **15**.

With reference to FIG. 2, one of the plurality of row units **30** is shown in greater detail. The row units **30** may be identical to each other in some implementations but may have some differences in other implementations. Each row unit **30** is provided with a central frame member **40** supported on the planter frame **25** by downforce control actuators. The central frame member **40** is configured to support a seeding system **45**, a press wheel **50**, and a mini hopper **55**. The seeding system **45** is operable to create a hole **60** in the field **20** or ground for receiving a sentinel seed **65** (FIG. 7) and to inject the sentinel seed **65** into the space. The press wheel **50** then fills in the space with dirt. The seeding system **45** is configured to create a plurality of the holes **60** in series, the holes **60** being discrete from each other. The discrete holes **60** are in contrast with other planters configured to create trenches. The seeding system **45** includes a planting assembly **70**, a driving assembly **75**, and a support assembly **77**.

In the illustrated implementation, the row unit **30** includes a single seeding system **45**. However, in some cases it may be desirable to include a second seeding system **45** on some or all of the row units **30**. The second seeding system **45** would be mounted to the central frame member **40** in line with the first seeding system **45** (e.g., such that both seeding systems **45** are configured to plant a single row). The second seeding system **45** may include its own driving assembly **70** or may be connected to all or part of the driving assembly **70** of the first seeding system **45**. In yet other implementations, each row unit **30** may include three or more seeding systems **45**.

As best illustrated in FIGS. 7-10, the planting assembly **70** includes a seed guide **80** and a spike **85**. The seed guide **80** is formed as a hollow tube **90** having a guide first end **95** and guide second end **100** (FIG. 3). The guide first end **95** is selectively disposed adjacent the ground **20** and the guide second end **100** is connected to the driving assembly **75**. The hollow tube **90** includes a tube wall **105** having a seed aperture **110** disposed between the guide first end **95** and the guide second end **100**. The seed aperture **110** may include a through-hole **115** disposed in the tube wall **105**. The seed guide **80** includes a guide channel **120** (FIG. 10) extending through the hollow tube **90**. The guide channel **120** may be formed as a through-hole defined by the tube wall **105**. The spike **85** is formed as a shaft **125** (such that the terms “spike” and “shaft” may be used interchangeably herein) with a sidewall **130** extending between a spike first end **135** (FIG. 9) and a spike second end **140** (FIG. 3). The sidewall **130** is elongated and extends parallel to reciprocating movement of the spike **85**. The spike first end **135** is tapered to a point **145** such that when inserted into the ground **20**, the first end creates the hole **60** (which may also be referred to herein as a discrete hole **60**), which may be tapered. The hole **60** may be a blind hole (such as a bore or a space or a pocket) in the ground **20** for receiving one of the sentinel seeds **65**. The spike second end **140** is coupled to the driving assembly **75**, as will be described in greater detail below. The spike **85** is slidably disposed within seed guide **80**, such that the guide channel **120** receives the shaft **125**. The spike **85** and seed

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guide **80** are both moveable independently with respect to the central frame member **40**. The spike **85** is movable along a stroke path with respect to the seed guide **80**. The spike **85** and the seed guide **80** are coaxial. In a first position (FIG. 7) along the stroke path, the spike first end **135** extends lower than the guide first end **95**. In the first position, the seed aperture **110** is blocked by the spike **85** such that the seed aperture **110** is not in fluid communication with the guide channel **120** of the seed guide **80**. In a second position (FIG. 9) along the stroke path, the spike first end **135** is disposed above the seed aperture **110** of the seed guide **80**. In the second position, the seed aperture **110** is open (not blocked by the spike **85**) such that the seed aperture **110** is in fluid communication with the guide channel **120** of the seed guide **80** (and ultimately in fluid communication with the hole **60** in the ground **20**). The spike **85** is movable through the stroke path with respect to the seed guide **80** from the first position to the second position and back to the first position, and so on, in a reciprocating fashion as illustrated in FIGS. 7-10. The stroke path may also include, in addition to the reciprocating component towards and away from the ground **20**, a rocking or tilting component, which will be described in greater detail below. (The terms “tilting” and “rocking” may be used interchangeably herein.) However, in some implementations, the stroke path may include only the reciprocating component.

A seed tube **150** and the seed guide **80** define the flow path that transports the sentinel seed **65** to the hole **60**. The seeding system **45** is configured to retain the sentinel seed **65** outside of the stroke path of the spike **85** until the spike **85** retracts a predetermined distance. The predetermined distance may be met when the spike **85** retracts past the seed aperture **110**. In some implementations, the predetermined distance may be met by other distances when the spike **85** retracts to other points along the stroke path.

With reference to FIG. 2, the driving assembly **75** includes a motor **155**, a first crankshaft **160**, and a second crankshaft **165**. The second crankshaft **165** is operably connected to the first crankshaft **160** by a timing belt **170**. The motor **155** may be directly coupled to the first crankshaft **160** or the second crankshaft **165** with the other driven by the timing belt **170**. In some implementations, the first crankshaft **160** and second crankshaft **165** may extend the length of a single row unit **30** and drive one planting assembly **70**. In other implementations, the first crankshaft **160** and second crankshaft **165** may extend across multiple row units **30** and drive multiple planting assemblies **70**. The first crankshaft **160** is coupled to a first radial link **175** having a first length R1. In the illustrated implementation, the first radial link **175** may include an arm mounted on the first crankshaft **160**. In other implementations, the first radial link **175** may include a post (not shown) mounted at a radius of R1 to a disc (not shown) mounted on the first crankshaft **160**. Other linkages are also contemplated. The first radial link **175** is pivotally coupled to a first rigid link **180**. The first rigid link **180** may be formed in a shape designed to avoid interference with other components of the row unit **30**, as illustrated, or may be formed as a straight bar in other implementations. The first rigid link **180** is rotatably coupled to the spike second end **140** of the planting assembly **70**. Similarly, the second crankshaft **165** is coupled to a second radial link **185** having a second length R2 which is pivotally coupled to a second rigid link **190**. The second rigid link **190** is coupled to the guide second end **100** of the seed guide **80** of the planting assembly **70** such that the seed guide **80** is aligned coaxially with the second rigid link **190**. In some implementations the seed guide **80** may be formed as a part

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of the second rigid link **190**. In other implementations, the driving assembly **75** may have other configurations for driving the spike **85** and the seed guide **80**, such as other arrangements of linkages coupled to a rotating shaft or shafts, or linear actuators (such as electric, pneumatic, or hydraulic linear actuators) may be employed. Other suitable mechanisms for driving the spike **85** and seed guide **80** are also contemplated.

The support assembly **77** supports the planting assembly **70** adjacent the ground **20**. The planting assembly **70** is supported by the support assembly **77** such that the stroke path of the spike **85** is oriented such that the spike **85** reciprocates extending towards and retracting away from the ground **20** and the sidewall **130** extends parallel to the reciprocation direction of the spike **85**. The support assembly **77** includes a flexible arm **195** extending generally vertically below the central frame member **40** and a support arm **200** coupled to a lower end of the flexible arm **195** and extending generally horizontally towards the planting assembly **70**. (The term “generally” is used herein to encompass the ability to pivot, move, and/or flex from the main orientation as described below and to indicate that the main orientation need not be exact, just approximate.) The support arm **200** is coupled to the seed guide **80** of the planting assembly **70** and provides support to the planting assembly **70** and driving assembly **75** to inhibit undesired horizontal motion. The support arm **200** supports the seed guide **80** such that the seed guide **80** may reciprocate independently from the spike **85**. The horizontal support arm **200** may be spring biased toward an intermediate position but may be able to extend or retract to allow the planting assembly **70** to deflect horizontally. The flexible arm **195** is formed from a material, such as a metal (such as steel, aluminum, or other metal), polymer, or composite (such as fiberglass), that is elastic enough to bend when a force is applied and resilient enough to return to the original form when the force is released. In other implementations, the support assembly **77** may include a single rigid bar coupling the seed guide **80** to the center frame member **40**. In other implementations, the seed guide **80** is directly coupled to the central frame member **40** via a pin joint or other pivoting connection. In other implementations, the support assembly **77** may include a rigid vertical support link extending from the central frame member **40** and a horizontal support arm extending between the support link and the seed guide **80**. The support assembly **77** may have other suitable configurations in other implementations.

The press wheel **50** includes an outer surface **205** and is configured to apply a downward force to the ground **20**. The press wheel **50** is supported on the central frame member **40** by a depth control actuator **210**. The depth control actuator **210** is actuatable to move the press wheel **50** vertically lower with respect to the central frame member **40** when the depth control actuator **210** is extended or vertically higher with respect to the central frame member **40** when the depth control actuator **210** is retracted. In the illustrated implementation, the press wheel **50** is supported on a linkage **215** that is rotated counterclockwise when the depth control actuator **210** is extended and clockwise when the depth control actuator **210** is retracted. Thus, the press wheel **50** has at least a component of vertical movement in response to actuation of the depth control actuator **210**. The rotation of the linkage **215** causes the vertical displacement of the press wheel **50** with respect to the central frame member **40**. The depth control actuator **210** may include a gas spring (not shown) controlled by gage wheels (not shown). In other implementations, the depth control actuator **210** may be

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pneumatically or hydraulically controlled. The press wheel **50** is rotated in response to a friction force between the ground **20** and the outer surface **205** when the planter **10** is moving. In other implementations, the press wheel **50** may be mounted to the central frame member **40** in other ways.

The central frame member **40** supports the mini hopper **55** adjacent the seeding system **45**. The mini hopper **55** is connected to the one or more hoppers **35** and receives sentinel seed **65** therefrom. The central frame member **40** also supports a seed meter **220** such as is known in the art to provide a regulated flow of seeds. The seed meter **220** receives sentinel seeds **65** from the mini hopper **55**, separates the sentinel seeds **65**, and supplies metered sentinel seeds **65** to the seed tube (which may also be referred to herein as a seed conduit), one after another in substantially equal sequential intervals. In some implementations, the seed meter **220** may receive sentinel seeds **65** directly from one of the hoppers **35**. The seed tube **150** may be pressurized by a pressure source **230** (e.g., a pump or other source of pressurized air) and is in fluid communication with the seed aperture **110** in the seed guide **80** (and thus fluidly connected to the guide channel **120** and to the hole **60** in the ground **20**). For example, the seed guide **80** may support an end of the seed tube **150** extending at least partially into the seed aperture **110**, or the seed tube **150** and the seed guide **80** may be coupled in any other suitable manner, e.g., using a connector between the seed tube **150** and the seed guide **80** in fluid communication with the seed aperture **110**. The pressure may be constant or may be timed to match the operation of the driving assembly **75**. In other implementations, the seed tube **150** need not be pressurized, e.g., the seed tube **150** may transport the seeds by gravity. In yet other implementations, the seed tube **150** may include a continuous loop belt for transporting the seeds to the seed guide **80**. In yet other implementations, the seed tube **150** may include a mechanism to mechanically urge the seeds out of the seed tube **150**. The seed tube **150** may form a first portion of a flow path configured to transport the seeds from the seed meter **220** towards the ground **20**. The seed guide **80** may define a second portion of the flow path, as will be described in greater detail below.

With reference to FIGS. 3-6, in operation, the vehicle **15** is operated to move the planter **10** across the ground **20** in a first direction **235** and at a planter speed **S1**. Each seeding system **45** may be started simultaneously or at different times. When started, the motor **155** of the row unit **30** is activated in order to actuate the driving assembly **75**. The activation of the motor **155** is separate from the movement of the planter **10**. Because of this, a seeding speed **S2** of the seeding system **45**, as measured by the number of seeds planted per unit time, is independent from the planter speed **S1**, as measured by the distance traveled per unit time. The activated motor **155** rotates the first crankshaft **160** counterclockwise (as viewed in FIGS. 3-6). As the first crankshaft **160** rotates, the timing belt **170** transfers the motion to the second crankshaft **165**, causing the second crankshaft **165** to rotate counterclockwise. The rotation of the first crankshaft **160** and second crankshaft **165** cause the driving assembly **75** to cycle through a plurality of stages. At the start of the cycle, shown in FIG. 3, the first radial link **175** is disposed pointing to the left of the first crankshaft **160** and the second radial link **185** is disposed pointing upward of the second crankshaft **165**. The second rigid link **190** extends generally vertically from the second radial link **185** to the seed guide **80**, which also is oriented generally vertically. The spike **85** is pivotally connected to the first rigid link **180** such that it follows the orientation of the seed guide **80**. The

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spike 85 is disposed between the first position and the second position relative to the seed guide 80. Both the spike first end 135 and the guide first end 95 are disposed above the ground 20. As shown in FIG. 4, the movement of the first crankshaft 160 and second crankshaft 165 rotates the first radial link 175 and second radial link 185, respectively. The first radial link 175 is disposed downward of the first crankshaft 160, moving the first rigid link 180 and the spike 85 down as well. The second radial link 185 is disposed left of the second crankshaft 165, moving the second rigid link 190 and seed guide 80 down and tilting each toward the left (as illustrated in FIG. 4). The spike 85 is now in the first position relative to the seed guide 80. The spike 85 is in the ground 20 (meaning engaged with and displacing soil to create the hole 60) and the seed guide 80 is adjacent the ground 20. The cycle continues to FIG. 5 where the first radial link 175 points right and the second radial link 185 points down. The seed guide 80 is oriented generally vertically, and the spike 85 is disposed between the first and second positions. The seed guide 80 is now in the ground 20, further displacing soil to create the hole 60. In other words, the seed guide 80 is configured to penetrate the ground 20 to create a portion of the hole 60. The rotation continues to the stage shown in FIG. 6 and the first radial link 175 points upward and the second radial link 185 points left. The seed guide 80 and spike 85 are tilted to the right and the spike 85 is in the second position. The seed guide 80 is adjacent the ground 20. The cycle then continues with the first radial link 175 returning to the left of the first crankshaft 160 and the second radial link 185 returning to the top of the second crankshaft 165 as shown in FIG. 3. The resultant motion of the planting assembly 70 in response to the driving assembly 75 is that the seed guide 80 and the spike 85 reciprocate independently with respect to each other and with respect to the central frame member 40. In other words, the seed guide 80 is configured to reciprocate asynchronously with respect to the spike 85. In some portions of the resultant motion, the spike 85 retracts while the seed guide 80 extends (FIG. 8) and in other portions of the resultant motion, the spike 85 extends while the seed guide 80 retracts (FIG. 10).

The movement of the planting assembly 70 is shown in more detail in FIGS. 7-10. As the planter 10 travels along the row in the first direction 235 and the driving assembly 75 operates the spike 85 and seed guide 80 both move toward the ground 20, as shown in FIG. 7. As the spike 85 pierces the ground 20, the tapered point 145 displaces the soil with minimal force. The spike 85 is tilted away from the first direction 235 as it pierces the ground 20, meaning the spike first end 135 is further in the first direction 235 than the spike second end 140. At this time a sentinel seed 65 from the mini hopper 55 has been supplied to the seed tube 150 by the seed meter 220 and travels down toward the planting assembly 70. As shown in FIG. 8, once the spike 85 has reached a maximum extension, it begins to move back toward the central frame member 40, leaving the hole 60 in the ground 20. The maximum extension is determined by the length R1 of the first radial link 175. A larger length R1 creates a maximum extension that extends deeper into the ground 20. The spike 85 is still tilted away from the first direction 235 as it travels out of the ground 20. The seed guide 80 continues to move downward toward a maximum extension and into the ground 20, displacing the soil further, still with minimal force required. The maximum extension of the seed guide 80 is determined by the length R2 of the second radial link 185, in a similar manner to the maximum extension of the spike 85. In other words, the maximum extension of the seed guide 80 defines an extended position for engaging the

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ground 20. At this point, the sentinel seed 65 is disposed adjacent the seed aperture 110 and is inhibited from entering the guide channel 120 by the sidewall 130 of the spike 85, the sidewall 130 acting as a barrier between the sentinel seed 65 and the hole 60. As shown in FIG. 9, the spike 85 retracts, e.g., follows the stroke path away from the ground 20. As the spike 85 retracts past the seed aperture 110, gravity and the pressure from the seed tube 150 pushes the sentinel seed 65 into the guide channel 120 of the seed guide 80. In some implementations, gravity may push the sentinel seed 65 into the guide channel 120 without the use of pressure. The spike 85 and the seed guide 80 are configured such that the sentinel seed 65 is blown past the spike 85 upon retraction of the spike past the seed aperture 110. The seed aperture 110 is disposed outside of the stroke path of the spike 85 as the seed aperture 110 is disposed in the tube wall 105 of the seed guide 80. In other words, the spike 85 does not enter the seed aperture 110.

The point 145 of the spike 85 is configured to not contact the sentinel seed 65 during shaft extension. However, the sidewall 130 may contact the sentinel seed 65 to block the sentinel seed 65, as described above, in some instances. The spike 85 is configured to not push the sentinel seed 65 into the ground 20. In some implementations, the spike 85 does not include a channel for transporting the sentinel seed 65 through the spike 85; rather, the sentinel seed 65 is transported past the spike 85. Damage to the sentinel seed 65 is inhibited by reducing contact between the spike 85 and the sentinel seed 65.

As shown in FIG. 10, the seed guide 80 has begun to move back toward the central frame member 40, tilted toward the first direction 235, having widened the hole 60, however the guide first end 95 is still in the ground 20. The sentinel seed 65, therefore, has followed the guide channel 120 down into the hole 60. Misplacement and damage of the sentinel seed 65 is inhibited by injecting the sentinel seed 65 when the guide first end 95 is still in the ground 20. As shown in FIG. 10, as the seed guide 80 moves toward the central frame member 40, away from the ground 20, the spike 85 begins to move back toward the ground 20, and the planting assembly 70 begins to tilt away from the first direction 235 once more. Because of the tilting of the planting assembly 70, the resultant hole 60 is minimally elongated in the first direction 235. The process repeats to create further holes 60 in series in the first direction 235. The planting assembly 45 is configured to create one hole 60 on each stroke.

The press wheel 50 (FIG. 2) follows the planting unit and applies a force that redistributes the soil, filling in the holes 60. The operation leaves a series of loose soil spots, in contrast to the loose trenches created by typical furrow disks. The press wheel 50 may be supported at different distances from the central frame member 40 in order to create different planting heights. The position of the press wheel 50 may be adjusted by actuating the depth control actuator 210 through the associated gage wheels. Extending the depth control actuator 210 lowers the press wheel 50 toward the ground 20, creating a shallower planting depth, and retracting the depth control actuator 210 raises the press wheel 50 away from the ground 20, creating a deeper planting depth.

In certain cases, the ground 20 is not uniform. For example, there may be rocks or hard spots embedded in the ground 20. As the spike 85 punches into the ground 20, it may run into the hard spot. To inhibit damage to the seeding system 45 and to reduce disruptions to the operation, the force of the rock R is deflected through the flexible arm 195. As illustrated in FIGS. 11-14, when the spike 85 hits a hard

spot, the flexible arm **195** may bend allowing the downward force provided by the driving assembly **75** to continue and the planting assembly **70** to move. In the illustrated implementation the flexible arm **195** bends such that a lower end of the flexible arm **195** moves away from the first direction **235** and the planting assembly **70** is horizontally displaced away from the first direction **235**. However, the flexible arm **195** and the planting assembly **70** may also deflect toward the first direction **235**. The direction of deflection of the flexible arm **195** and the planting assembly **70** may be based on the geometry of the hard spot and the reaction forces within the planting assembly **70**.

The planter design as described above requires less down force than typical furrowing discs. Because the planter **10** has to support less down force, the central frame member **40** can be made of lighter materials. The seeding system **45** design also makes it easy to synchronize the seeding across the rows, making it easy to create a grid planted field. The holes created by the seeding system **45** are relatively small and therefore create less loose soil which makes erosion and subsequent seed loss less likely. Additionally, the driving assembly **75** is driven independently of the motion of the planter **10**. This allows seeding speed **S2** to be varied independently of the planter speed **S1**, allowing for more precise seeding. The small profile of the seeding system **45** and the holes created by the seeding system **45** allows for less distance between row units **30** and therefore less distance between rows. Thus, more rows can be planted.

The planter **10** is operable to simultaneously plant seeds of two or more different plant genotypes in a single planting operation, or in a single pass of the planter **10** through the field **20**. Specifically, the hoppers **35** may each be filled with different seeds of different plant genotypes. Seeds of different plant "genotypes" can include without limitation seeds of different plant varieties, of different plant cultivars, of different plant hybrids, of different plant species, or of different plant characteristics. The different seeds may be distributed to different row units **30** of the planter **10**, or to different seeding systems **45** of a single row unit **30**, or different seeds may be selectively supplied to a single seeding system **45**, as will be further described below. As such, the planter **10** can be utilized to practice intercropping, a multiple cropping practice where two or more crops are grown in proximity. Intercrops may include a first crop, or cash crop, of a first plant type to be harvested, and may further include one or more cover crops of different plant types that are planted between rows of the cash crop or amongst the cash crop. The one or more cover crops can be selected to manage attributes of the soil, such as soil erosion, soil fertility, soil quality, water, weeds, pests, diseases, biodiversity, wildlife, and other attributes.

Additionally, the planter **10** may be used to plant sentinel seeds **65** and other seeds in proximity of, or near, the sentinel seeds **65**. Sentinel seeds **65** exhibit different characteristics when presented with a stressor **240** (FIG. **15**). With continued reference to FIG. **15**, the different characteristics can be any detectable plant attribute that is proportional to the stressor **240** such as an abiotic factor **245** or a biotic factor **250**. For example, the abiotic factor **245** may be a lack of or an abundance of water, humidity, light or minerals. The abiotic factor **245** may also be soil conditions or temperature. The biotic factor **250** may be the presence of a certain fungus, bacteria, or insect.

For example, the sentinel seeds **65** may be genetically modified corn or soy beans that exhibit different characteristics from the other corn or soy beans that the sentinel seeds **65** are planted with, respectively. In other examples, the seed

may be traditionally bred. The first planting system **12** may be configured for planting a first sentinel plant **255** that exhibits a first sentinel plant characteristic **260**. The first sentinel plant **255** may be a sentinel seed **65** or a sentinel seedling (not shown) or any other sentinel plant form. The first sentinel plant **255** may also be a fungus, a lichen, mold, or any other stationary life form. The second planting system **14** may be configured for planting a second sentinel plant **280** that exhibits a second sentinel plant characteristic **285**. The second sentinel plant characteristic **285** may be the same as the first sentinel plant characteristic **260** or it may vary. The second sentinel plant **280** may be a sentinel seed **65** or a sentinel seedling (not shown) or any other sentinel plant form. The second sentinel plant **280** may also be a fungus, a lichen, mold, or any other stationary life form.

In one embodiment, the first or second sentinel plant characteristic **260**, **285** may include at least one of varying intensity of electromagnetic response **265** generated by the first or second sentinel plant **255**, **280** or a plant growth rate **270** that correspond to a magnitude **275** of the stressor **240**. The electromagnetic response **265** may be an absorption, transmission, backscatter, reflectance, fluorescence, bioluminescence, or other. The electromagnetic response **265** may be induced with or without external stimulation.

In another embodiment, the first or second sentinel plant characteristic **260**, **285** may include a change in a leaf feature **225** such as a change in leaf size, color, color pattern, shape, area index, or temperature. Other leaf features **225** are contemplated by this disclosure.

In yet another embodiment, the first or second sentinel plant characteristic **260**, **285** may include a change in a plant stem feature **370** such as a change in plant stem height, plant stem biomass, plant stem formations or malformations, plant stem diameter, plant stem color, or plant stem color pattern. Other plant stem features **370** are contemplated by this disclosure.

The first or second sentinel plant characteristic **260**, **285** may include the slow plant growth rate **270** or otherwise altered growth rate, or a time to emergence, seed germination rate, or an initiation of reproductive phase.

In another embodiment, the first or second sentinel plant characteristic **260**, **285** may include a change in a flower feature **375** such as flower color or flower timing. Other flower features **375** are contemplated by this disclosure.

In yet another embodiment, the first or second sentinel plant characteristic **260**, **285** may include a senescence feature **380** such as death of the sentinel plant due to daylight, a daylight change derivative, a change in temperature, a soil chemical attribute, or other. Other first or second sentinel plant characteristics **260**, **285** are contemplated by this disclosure.

Referring to FIG. **15**, a block diagram is provided of one example of a computing architecture **290** that includes planter **10**, a sensor **295**, a global positioning system ("GPS") **300**, and a control system **305**. Global positioning system **300** may comprise a Global Navigation Satellite System (GNSS), a terrestrial radio triangulation system, or any other system which is able to provide the location of the planter **10** in the field **20** in global or local coordinates. FIG. **15** illustratively shows that the planter **10**, the sensor **295**, the GPS **300**, and the control system **305** are connected over a network **310**. Thus, computing architecture **290** operates in a networked environment, where the network **310** includes any of a wide variety of different logical connections such as a local area network (LAN), wide area network (WAN), controller area network (CAN) near field communication network, satellite communication network, cellular net-

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works, or a wide variety of other networks or combination of networks. It is also noted that the control system **305** can be deployed on the planter **10** such that the control system **305** performs the operations described herein without a networked connection. In addition, while the present description will primarily focus on an example of the control system **305** communicating with the planter **10**, it is noted that the same or similar functionality can be provided when communicating with a wide variety of vehicles **15** and/or remote systems.

The sensor **295** may be communicatively coupled to the planter **10**. For example, the sensor **295** may be coupled to the planter **10** or it may be coupled to a satellite **312** that communicates with the planter **10** or control system **305**. The sensor **295** may be configured for sensing a current field characteristic **315** and generating a current field characteristic signal **320** indicative of the current field characteristic **315** that can be used to determine where to plant the sentinel seeds **65**. The sensor **295** may include an electromagnetic sensor **325** (e.g., optical sensor) and the current field characteristic **315** may include an image **330** that depicts a field condition **335**. The image **330** may be 2D or 3D. The image **330** may comprise a single pixel or data value. The electromagnetic sensor **325** may be a camera or other imaging sensor such as a video camera. Alternatively, the sensor **295** may include at least one of a radar, lidar, laser-based sensors, LIDAR based sensors, temperature sensors, soil property sensors, and a wide variety of other imaging or other sensing systems.

The current field characteristic **315** may include an image **330** that depicts the field condition **335** such as a topography **385** of the field **20** or landscape position **390** of plants **11** or a size of the plant **11** patches. The current field characteristic **315** may also include soil type **395** such as clay composition or top soil composition. The current field characteristic **315** may include a moisture level determination **400** such as 20-80% water content or a nutrient level determination **405** that includes a level of soil organic matter, or residue cover, or cover crops. Additionally, the current field characteristic **315** may include a temperature reading **410** of the field **20** at a soil depth or a pH reading **415** of the soil of the field **20**.

The current field characteristic **315** may include a compaction reading **420** of the field **20** that includes compaction layer details such as depth of the compaction and compaction drainage and its location in the field **20**. Additionally, the current field characteristic **315** may include a weather reading **425**. The weather reading **425** may include past, present, and future temperatures, moisture, solar, sun angle, sunlight attenuation by clouds or other, day length, or other.

The current field characteristic **315** may include a field operations reading **430** that indicates machinery data including machinery settings or timing of machinery operations along with the location in the field **20**. Additionally, the current field characteristic **315** may include a crop reading **435** that indicates crop details including mono or multiple varieties that are planted in the field **20** and their location. The current field characteristic **315** may include a pests reading **440** including an indication of any insects, mammals, birds, or other similar biotic factors and their location in the field **20**. The current field characteristic **315** may include a disease reading **445** including an indication of any bacteria, molds, smuts, viruses or other similar biotic factors and their location in the field. The current field characteristic **315** may include a weed reading **450** including an indication of any weeds and weed seeds and their location in the field **20**. Other current field characteristics **315** are contemplated by this disclosure.

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The GPS **300** is communicatively coupled to the planter **10**. The GPS **300** is configured for generating a location signal **340** indicative of a location **345** of the planter **10** or vehicle **15**. Generally, the GPS **300** receives sensor signals from one or more sensors, such as a GPS receiver, a dead reckoning system, a LORAN system, or a wide variety of other systems or sensors, to determine the location **345** of the planter **10** across the field **20**.

The control system **305** is in communication with the sensor **295** and the GPS **300**. The control system **305** is communicatively coupled to the planter **10**. The control system **305** may be configured to receive the first sentinel plant characteristic **260**, receive the second sentinel plant characteristic **285**, receive the location signal **340**, receive a georeferenced field characteristic **350** from a data storage **355**, and control at least one of the first planting system **12** or the second planting system **14** based on at least one of the first sentinel plant characteristic **260**, the second sentinel plant characteristic **285**, the location **345**, the current field characteristic **315**, or the georeferenced field characteristic **350**, or any combination thereof. The georeferenced field characteristic **350** may include a depth and spacing information for the plants **11** from a prior planting operation or historical data for the field **20**. The georeferenced field characteristic **350** may include relationships between first and second sentinel plant characteristics **260**, **285** that can be sensed and the georeferenced field characteristics **350** that promote a degree of response.

The control system **305** may be configured to control the first planting system **12** to plant the first sentinel plant **255** at the location where the georeferenced field characteristic **350** indicates the stressor **240** for the first sentinel plant characteristic **260**. For example, the georeferenced field characteristic **350** may indicate an area or location of the field **20** where there is or may be a high incidence of an insect. The control system **305** may control the first planting system **12** to plant the first sentinel plant **255** that has the first sentinel plant characteristic **260** that is elicited by the insect at this location. The first sentinel plant characteristic **260** may be the slow growth rate **270**.

In an additional example, the georeferenced field characteristic **350** may indicate an area of the field **20** where there is a high probability of soybean chlorosis in some years. The control system **305** may control the first planting system **12** to plant the first sentinel plant **255** that has the first sentinel plant characteristic **260** that is elicited by difficulties in plant uptake of iron, resulting in chlorosis, at this location. The first sentinel plant characteristic **260** may be a yellowing of leaves proportional to the severity of the stress. Mitigation could be to apply chelated iron and maybe sulfur with the planter **10**, the vehicle **15**, or other equipment.

The georeferenced field characteristic **350** may include a sentinel plant map **360** and the control system **305** may be configured to update the sentinel plant map **360** and the georeferenced field characteristic **350** in the data storage **355** with the current field characteristic **315**. The sensor **295** may be used to obtain the data stored in the data storage **355**. The data storage **355** may be coupled to the planter **10**, the work vehicle **15**, or the satellite **312**. In some examples, satellite **312** may comprise a manned aircraft or unmanned aerial vehicle as a remote sensor platform.

A display **365** may be provided for displaying the sentinel plant map **360** to an operator. Additionally, the operator may enter planter **10** or vehicle **15** commands via the display **365**. Display **365** may include audio elements such as a speaker or microphone. Display may include haptic elements such as a vibration generator.

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Referring now to FIG. 16, a flow diagram of a method 500 for controlling a planter 10 on a field 20 for planting a sentinel plant 11 is provided. At 505, a first sentinel plant characteristic 260 is received. At 510, a location signal 340 indicative of a location 345 of the multi-variety planter 10 is received. At 515, a georeferenced field characteristic 350 is received.

At 520, a first planting system 12 is controlled to plant the first sentinel plant 255 based on at least one of the first sentinel plant characteristic 260, the location 345, or the georeferenced field characteristic 350. Alternatively, a current field characteristic 315 is sensed and the first planting system 12 is controlled to plant the first sentinel plant 255 based on the first sentinel plant characteristic 260, the location 345, and at least one of the georeferenced field characteristic 350 or the current field characteristic 315. Alternatively, the first planting system 12 may be controlled to plant the first sentinel plant 255 at the location where the georeferenced field characteristic 350 indicates a stressor 240 that may or will cause the first sentinel plant 255 to exhibit the first sentinel plant characteristic 260. In other examples, the first planting system 12 may be controlled to plant the first sentinel plant 255 at a location which is part of a geographic pattern such as a grid, an elevation gradient, a transect, or other geospatially-dependent scheme.

At 525, a prescriptive sentinel plant map 535 may be generated. The prescriptive sentinel plant map 535 may include a latitude, a longitude, and a depth of where every plant 11 should be planted in the field 20. The prescriptive sentinel plant map 535 may be used by the planter 10 to plant plants 11 including sentinel seeds 65.

At 530, the sentinel plant map 360 may be displayed to an operator. The sentinel plant map 360 may be updated to include as-planted data such as where every plant 11 was planted or where and what type of sentinel seeds 65 were planted.

Advantageously, sentinel seeds 65 would not need to be planted at the same time as the main crop. They could be planted at the time of spring tillage ahead of the crop planting pass. They could also be planted after the main crop. In one example, a problem is otherwise identified in the field 20 and then sentinel seeds 65 are planted to play a diagnostic role to find out why the stand is poor or lagging. In another example, sentinel seeds 65 could be planted just prior to a critical crop phase such as pollination to provide data on conditions during a window. Sentinel seeds 65 could also be perennials planted once for multiple seasons of use for annual crops or multi-year crops like cane sugar.

Sentinel seeds 65 could be an organism, other than plants 11, such as bacteria or fungi (some soil fungi in the tropics are naturally fluorescent). The response of the sentinel organisms could be detected by sensors 295 on a ground engaging element such as a shank. Similarly, the response to a stressor 240 by a sentinel seed 65 may be in the roots and observed below ground.

Sentinel seeds 65 could depend on allelopathy. A first plant, micro, fungi could be the one which is proportionally responsive to the stressor 240. The response may be in the form of proportionally altering the environment around a second plant. The second plant then communicates the level of that alternate environment. For example, in response to a soil condition, a microbe alters the pH of the soil adjacent to it. This causes a color change in the flower of a co-located plant sensitive to soil pH. A liquid may be precisely applied in the seed trench and planting a sentinel seed 65 may comprise placing a seed and inoculating the soil near the seed with bacteria, fungi, etc. The inoculant wouldn't even

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need to be alive. It could be the first plant is replaced by a chemical that reacts with one or more soil components, living or non-living, that results in a proportional amount that is reported via the second plant. Chemical may be applied as a liquid, solid, or gas. Also note that the reporter plant response can be other things besides fluorescence such as a nitrogen level experienced by a microbe, a local soil pH level, or a plant flower or leaf color.

Machine learning may be used to improve the efficacy of where to plant plants 11 including sentinel seeds 65. The ability of the sentinel plants to accurately sense desired crop conditions, in some cases compared to laboratory analysis or other sensing means, may be improved with machine learning. This improvement could be in the selection of varieties to use as sentinels, planting attributes (e.g., interplant spacing, depth), or location of individual plants 11 or plant 11 patches.

What is claimed is:

1. A planter configured for planting a sentinel plant in a field, the planter comprising:

- a first planting system coupled to the planter, the first planting system configured for planting a first sentinel plant having a first sentinel plant characteristic;
- a global positioning system communicatively coupled to the planter, the global positioning system configured for generating a location signal indicative of a location of the planter;
- a data storage for storing a georeferenced field characteristic, the georeferenced field characteristic including historical data for the field;
- a sensor communicatively coupled to the planter, the sensor configured for sensing a current field characteristic and generating a current field characteristic signal indicative of the current field characteristic; and
- a control system communicatively coupled to the planter, the control system configured to:
 - receive the first sentinel plant characteristic,
 - receive the location signal indicative of the location of the planter,
 - receive the georeferenced field characteristic,
 - receive the current field characteristic, and
 - control the first planting system based on the first sentinel plant characteristic, the location, and at least one of the georeferenced field characteristic and the current field characteristic to plant the first sentinel plant at the location that includes a stressor that elicits the first sentinel plant characteristic.

2. The planter of claim 1, further comprising a second planting system coupled to the planter, the second planting system configured for planting a second sentinel plant having a second sentinel plant characteristic, the control system configured to receive the second sentinel plant characteristic, control at least one of the first planting system or the second planting system based on at least one of the first sentinel plant characteristic, the second sentinel plant characteristic, the location, and at least one of the georeferenced field characteristic and the current field characteristic to plant the first sentinel plant or the second sentinel plant at the location that includes a stressor that elicits the first sentinel plant characteristic or the second sentinel plant characteristic.

3. The planter of claim 1, wherein the georeferenced field characteristic comprises a sentinel plant map and the control system is configured to update the sentinel plant map with the current field characteristic.

4. The planter of claim 3, further comprising a display for displaying the sentinel plant map to an operator.

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5. The planter of claim 1, wherein the first sentinel plant characteristic comprises at least one of varying intensities of electromagnetic response generated by a sentinel plant or a plant growth rate that correspond to a pressure of the stressor.

6. A multi-variety planter configured for planting a sentinel plant in a field, the multi-variety planter comprising:

- a first planting system coupled to the multi-variety planter, the first planting system configured for planting a first sentinel plant having a first sentinel plant characteristic;
- a second planting system coupled to the multi-variety planter, the second planting system configured for planting a second sentinel plant having a second sentinel plant characteristic;
- a global positioning system communicatively coupled to the multi-variety planter, the global positioning system configured for generating a location signal indicative of a location of the multi-variety planter;
- a data storage for storing a georeferenced field characteristic, the georeferenced field characteristic includes historical data for the field;
- a sensor communicatively coupled to the planter, the sensor configured for sensing a current field characteristic and generating a current field characteristic signal indicative of the current field characteristic; and
- a control system communicatively coupled to the multi-variety planter, the control system configured to:
 - receive the first sentinel plant characteristic,
 - receive the second sentinel plant characteristic,
 - receive the location signal indicative of the location of the planter,
 - receive the georeferenced field characteristic,
 - receive the current field characteristic; and
 - control at least one of the first planting system or the second planting system based on the first sentinel plant characteristic, the second sentinel plant characteristic, the location, and at least one of the georeferenced field characteristic and the current field characteristic to plant the first sentinel plant or the second sentinel plant at the location that includes a

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stressor that elicits the first sentinel plant characteristic or the second sentinel plant characteristic.

7. The multi-variety planter of claim 6, wherein the sensor comprises an electromagnetic sensor, the current field characteristic comprises an image that depicts a field condition.

8. The multi-variety planter of claim 6, wherein the georeferenced field characteristic comprises a sentinel plant map and the control system is configured to update the sentinel plant map with the current field characteristic.

9. The multi-variety planter of claim 8, further comprising a display for displaying the sentinel plant map to an operator.

10. The multi-variety planter of claim 6, wherein the first sentinel plant characteristic comprises at least one of varying intensities of electromagnetic response generated by a sentinel plant or a plant growth rate that correspond to a pressure of the stressor.

11. The multi-variety planter of claim 10, wherein the stressor comprises at least one of an abiotic factor or a biotic factor.

12. A method of controlling a planter for planting a sentinel plant in a field, the method comprising:

- receiving a first sentinel plant characteristic;
- receiving a location signal indicative of a location of the planter;
- receiving a georeferenced field characteristic from a data storage, the georeferenced field characteristic including historical data for the field;
- receiving a current field characteristic; and
- controlling a first planting system to plant the first sentinel plant based on the first sentinel plant characteristic, the location, and at least one of the georeferenced field characteristic and the current field characteristic to plant the sentinel plant at the location that includes a stressor that elicits the first sentinel plant characteristic.

13. The method of claim 12, wherein the first sentinel plant characteristic comprises at least one of varying intensities of electromagnetic response generated by a sentinel plant or a plant growth rate that correspond to a pressure of the stressor.

14. The method of claim 12, further comprising displaying a sentinel plant map to an operator.

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